STUDENTS INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES)

UNDERTAKEN AT

INTERNATIONAL INSTITUTE OF TROPICAL AGRICULTURE (IITA)

TECHNICAL WORK REPORT

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March 2026

LETTER OF SUBMISSION

Department of Agricultural/Env Engineering, Faculty of Technology,

University of Ibadan, Ibadan.

March 25, 2026.

The Director,

Industrial Training Coordinating Centre, University of Ibadan,

Ibadan.

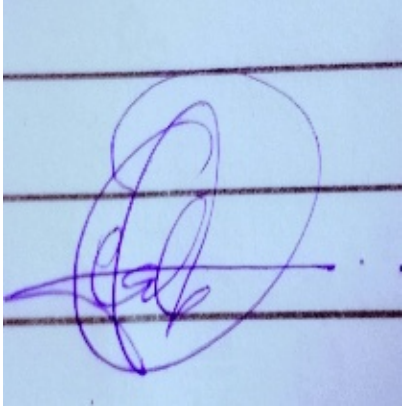
Dear Sir,

SUBMISSION OF INDUSTRIAL TRAINING REPORT

I, Olalere Oluwaferanmi Marvellous, a 300L student of the department of Agricultural and Environmental Engineering with matriculation number 237890, write this letter to notify you of the completion of my 3 months Students' Industrial Work Experience (SIWES), for the 2023/2024 academic session. The program commenced 5th January, 2026 and ended on the 13th March, 2026.

I am turning in this submission letter along with my industrial training report and logbook. I hope my work report is examined and reviewed favourably.

Yours faithfully,



Olalere O. Marvellous.

ACKNOWLEDGEMENT

I am grateful to Almighty God for guiding me through my industrial training. I also extend my appreciation to my parents for their unwavering support. The opportunity provided by the Industrial Training Coordinating Center (ITCC) at the University of Ibadan has been invaluable, offering me practical experience that has complemented my theoretical knowledge.

This training has been a pivotal moment in my engineering journey, providing insight and inspiration that will drive me to succeed. I am particularly thankful to Mr. James Fyanka, the supervisor in charge, for his supervision, guidance, and unwavering support throughout the program. His expertise and encouragement have been invaluable to my growth.

Additionally, I appreciate the staff, apprentices, and fellow engineers who have supported and motivated me during this training. Your collective efforts have made a significant impact on my engineering career.

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ABSTRACT

This report presents a detailed account of the knowledge and experience acquired during the Students Industrial Work Experience Scheme (SIWES) at the International Institute of Tropical Agriculture (IITA), Ibadan.

The training focused primarily on activities at the Biogas Unit, where organic waste materials such as cow dung and poultry manure were processed into usable biogas through anaerobic digestion. The report highlights practical involvement in slurry preparation, digester feeding, gas production, and compression processes.

It also explains the operational principles of key equipment such as the hammer mill, portable electric agitator, manometer, and flowmeter. Furthermore, concepts such as gas compression, pressure-volume relationships (Boyle's Law), and system efficiency were studied.

The training provided valuable hands-on experience, bridging the gap between theoretical knowledge and real-world engineering applications.

CHAPTER ONE

ORIGIN OF SIWES

The Students Industrial Work Experience Scheme (SIWES) was established in 1973 by the Industrial Training Fund (ITF) to provide tertiary institution students with practical industry experience aligned with their field of study. This structured program aims to develop competent and informed individuals by bridging the gap between theoretical knowledge and real-world applications (Made, 2009). The Federal Government funds SIWES, which is coordinated by the ITF and National Universities Commission (NUC).

OBJECTIVES OF SIWES

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    - Provide students with the opportunity to apply theoretical knowledge in real-world settings
    - Familiarize students with industry facilities, machinery, and professional work methods
    - Instill acceptable work ethics, teamwork, and accountability in students

SIGNIFICANCE OF SIWES IN TERTIARY EDUCATION

Nigeria's educational system has traditionally prioritized theoretical learning over practical applications. SIWES has successfully addressed this imbalance by offering undergraduates practical experience, enabling them to connect theoretical knowledge with real- world applications. This scheme enhances learning, fosters work ethics, teamwork, and accountability, preparing students for future professional endeavors.

CHAPTER TWO INTERNATIONAL INSTITUTE OF TROPICAL AGRICULTURE

HISTORY OF IITA

The International Institute of Tropical Agriculture (IITA), headquartered in Ibadan, Nigeria, is a renowned research institute dedicated to transforming African agriculture and ensuring food security. Founded in 1967, IITA is a member of the CGIAR system organization, a network of research institutes across Africa. As a non-profit organization, IITA collaborates with prominent organizations like UNICEF, FAO, and WHO to develop agricultural solutions, with funding from donors passionate about improving Africa's agricultural sector. IITA focuses on research and development of staple food crops, including banana, cassava, and maize, with state- of-the-art facilities like the bioscience center and genetic resource center.

International Institute of Tropical Agriculture

OBJECTIVES OF IITA

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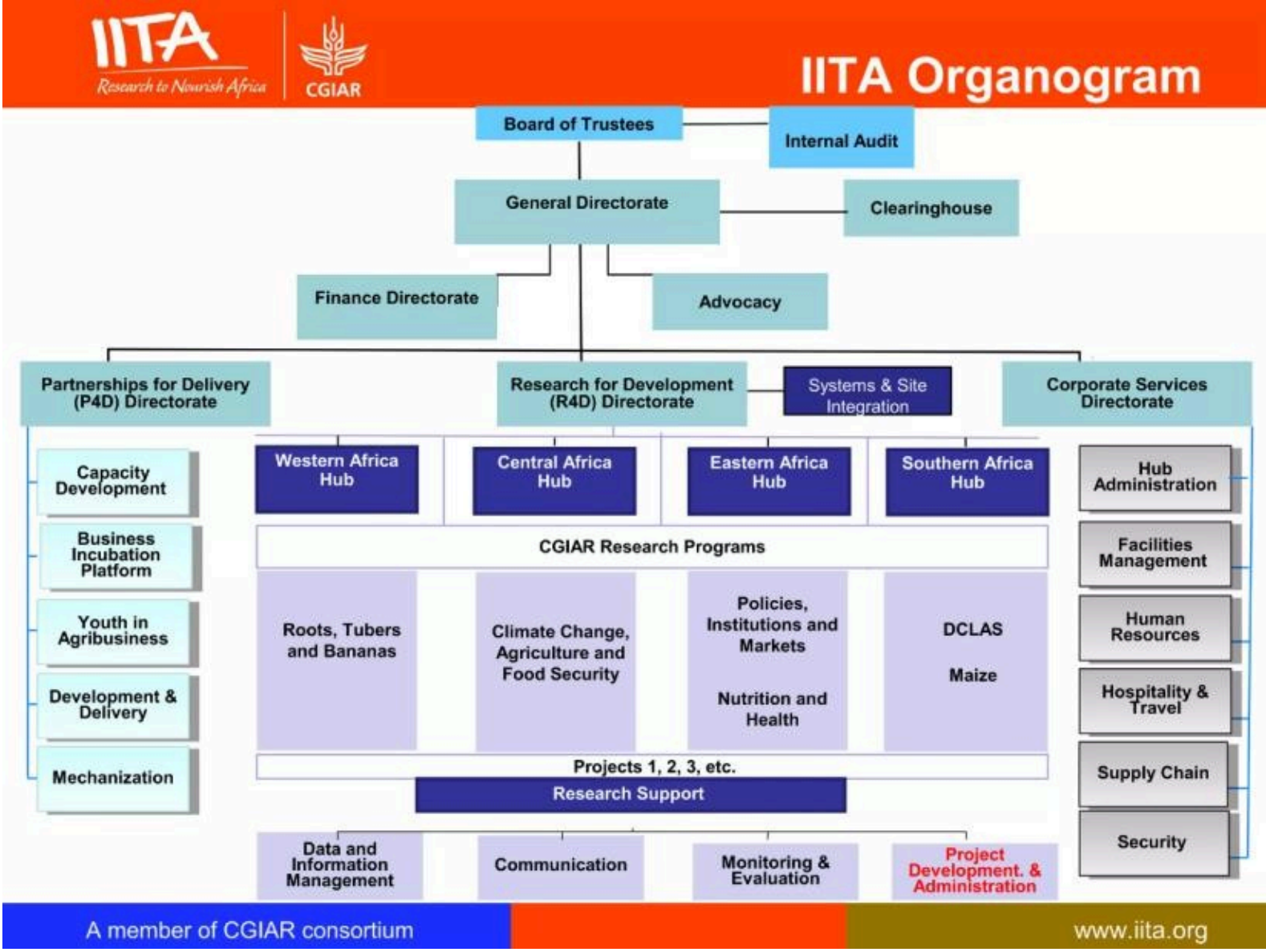
    - Develop sustainable natural resource management systems for agriculture (Atayi, 2001)
    - Improve nutritional quality of food products along crop value chains
    - Enhance breeding of nutritious staple crops in Africa
    - Support smallholder farmers in turning their farming activities into viable business operations
    - Conduct research on new genotypes of staple foods resistant to factors causing food unavailability

ORGANIZATIONAL STRUCTURE OF IITA

The Institute of Tropical Agriculture (IITA) in Ibadan is governed by an external Board of Trustees, which comprises the Director General, representatives from major donors, CGIAR, and the Nigerian government. This board oversees the institute's activities across Nigeria and Africa.



IITA Organogram

The day-to-day operations of IITA are managed by the Senior Management Team (SMT), led by the Director General and comprising Deputy Director Generals and various

directors, such as finance, research, and humid tropics. The SMT reports directly to the Board of Trustees.

The institute has three main directorates: Partnership for Delivery, Research for Development, and Corporate Services, each headed by a Deputy Director General. These directorates oversee various units, with scientists leading research units in different laboratories. The Corporate Services Directorate handles administrative services, while the Partnership for Delivery Directorate collaborates with external partners and donors to foster relationships and ensure accountability.

CHAPTER THREE

BIOGAS UNIT

The Biogas Unit is responsible for converting organic waste into biogas through anaerobic digestion. This process involves the breakdown of organic materials in the absence of oxygen to produce methane-rich gas used for energy.

The unit consists of:

- Two biodigesters (50 m^3 and 147 m^3 capacities)
- A gas holder (20 m^3 capacity)
- Gas compression system
- Storage tanks (A, B, and C)

Technical Overview: Biogas Production and Storage Unit

The Biogas Unit serves as the primary processing infrastructure for converting organic waste into renewable energy. Utilizing anaerobic digestion, the system breaks down biodegradable materials in a strictly controlled, oxygen-free environment to yield methane-rich biogas. The facility is engineered for staged biological processing, precise pressure regulation, and secure, high-capacity storage.

1. Anaerobic Digestion Infrastructure

Components: Two Biodigesters (50 m³ and 147 m³ capacities) **Functionality and Operation:** The biodigesters form the biological core of the unit. The presence of two distinct volumetric capacities suggests a staggered or two-stage digestion configuration, which optimizes the biochemical breakdown of waste:

- **Hydrolysis and Acidogenesis (Primary Stage):** The smaller 50 m³ digester is likely utilized for the initial phases of digestion. Here, complex organic polymers (carbohydrates, lipids, proteins) are broken down into simpler monomers and volatile fatty acids. This process is relatively fast and lowers the pH of the substrate.
- **Methanogenesis (Secondary Stage):** The larger 147 m³ digester accommodates the final, most sensitive stage of biogas production. Methanogenic bacteria, which require a strict pH balance and a longer hydraulic retention time, convert the acids into methane (CH_4) and carbon dioxide (CO_2). The larger volume allows for the extended processing time necessary to maximize the overall gas yield.

2. Gas Balancing and Pressure Regulation

Component: Gas Holder (20 m³ capacity) **Functionality and Operation:** Biogas generation is a dynamic biological process; production rates fluctuate naturally based on feedstock composition, ambient temperature, and microbial activity levels.

- **Pneumatic Buffering:** The 20 m³ gas holder acts as a critical pneumatic buffer between the biological and mechanical stages of the plant.
- **Pressure Stabilization:** It temporarily captures the raw, low-pressure biogas emerging from the digesters. By absorbing spikes in gas production, it stabilizes the internal pressure of the digesters and ensures a steady, regulated feed of gas into the downstream compression equipment.

3. Mechanical Pressurization

Component: Gas Compression System **Functionality and Operation:** Raw biogas exits the digestion phase at near-atmospheric pressure, which is entirely insufficient for compact storage, transport, or efficient downstream energy generation.

- **Energy Densification:** The gas compression system mechanically reduces the volume of the raw biogas, drastically increasing its pressure and energy density.
- **System Flow:** This mechanical lift is vital for overcoming pipeline resistance and pushing the gas into the final high-pressure storage infrastructure. In standard operations, this stage often works in tandem with moisture traps and desulphurization filters to protect the compressor from corrosion.

4. Redundant Storage System

Components: Storage Tanks A, B, and C **Functionality and Operation:** The terminal stage of the Biogas Unit involves capturing the compressed, energy-dense gas. Distributing the final storage across three distinct tanks provides critical operational advantages:

- **Operational Flexibility:** Segregated tanks allow the facility to route gas dynamically. One tank can be actively filling while another is discharging to a generator or pipeline.
- **Maintenance and Safety:** Multi-tank redundancy means that Tank A can be safely isolated and taken offline for routine maintenance, internal inspections, or pressure testing without halting the plant's continuous gas production, as Tanks B and C can absorb the load.

CHAPTER 5

PERSONAL EXPERIENCE, CHALLENGES & CONCLUSION

PERSONAL EXPERIENCE

My experience at the International Institute of Tropical Agriculture has been a defining moment in my pre-engineering journey. I had the opportunity to engage with talented individuals, learn from distinguished experts, and collaborate with forward-

thinking engineers who are driven to create a better world through innovative solutions and sustainable practices.

This experience has not only deepened my understanding of theoretical concepts but also provided me with hands-on experience, exposing me to the latest technologies and methodologies in precision farming. I have gained a new appreciation for the critical role that engineering plays in addressing global challenges, and I am now more determined than ever to pursue a career that combines my passion for engineering with my desire to make a positive impact on society.

My time at the institute has been a truly enriching experience, providing me with a unique perspective, essential skills, and a network of like-minded professionals who share my passion for innovation and excellence. I am confident that the lessons I have learned and the connections I have made will have a lasting impact on my future endeavors.

CHALLENGES FACED DURING THE SIWES PROGRAM

During my training at IITA, I encountered two main challenges. Firstly, the trainers were hesitant to provide us with adequate field experience, limiting our hands-on training to occasional minor tasks. Secondly, the vast distances between units made it difficult and exhausting to move around and explore different areas. Despite these challenges, I overall had a positive experience

CONCLUSION

In conclusion, my experience at the International Institute of Tropical Agriculture has been the highlight of my pre-engineering career. I had the privilege of connecting with exceptional minds and collaborating with engineers who share a vision to revolutionize the world through innovative problem-solving. This opportunity has helped me bridge the gap between theoretical knowledge and practical application.

Furthermore, I was exposed to the global aspects of engineering, and precision farming particularly caught my attention. I am now eager to pursue further research in this field and explore its potential to transform the agriculture industry

REFERENCES

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  -
    - Alam, A. (2006). Future requirements of agricultural machines for mechanizing agriculture. Status of farm mechanization in India. Report, Indian Council of Agricultural Research, India, 175-196.

DHARMARAJAN, V. (2022). DESIGN AND FABRICATION OF MULTI-PURPOSE AGRICULTURE MACHINE.

-
  -
    - Sethi, V. P., Dubey, R. K., & Dhath, A. S. (2009). Design and evaluation of modified screen net house for off-season vegetable raising in composite climate.
    - Energy conversion and management, 50(12), 3112-3128.
    - WAHAB, K. A. MICHAEL OKPARA UNIVERSITY OF AGRICULTURE UMUDIKE ABIA STATE COLLEGE OF ENGINEERING AND ENGINEERING TECHNOLOGY AGRICULTURAL AND BIORESOUCES ENGINEERING A PARTIAL FULFILMENT OF STUDENT INTERNSHIP WORKING EXPERIENCE (SIWES) FOR A PERIOD OF SIX MONTHS FROM MAY–OCTOBER 2018.